

SRS – DRUG TRACKING SYSTEM

Introduction

The purpose of this document is to define the functional and non-functional requirements of the **Drug Tracking System** to ensure a clear understanding.

The system will allow:

- Patients to search for available drugs.
- Pharmacies to assign available drugs.
- Administrators to monitor interactions on the platform.

Overall description

The Drug Tracking System is a web-based platform that allows patients to search for unavailable or hard-to-find drugs and identify pharmacies where they are currently available.

- Operating environment:
 - Web application.
 - Semi-structured database (MongoDB).

Functional requirements

- FR-1: The system shall allow users to log in using their email and password.
- FR-2: The system shall allow users to reset their passwords.
- FR-3: The system shall restrict access based on user role (patient, pharmacy, admin).
- FR-4: The system shall display a visible search bar on the home page without requiring scrolling.
- FR-5: After hitting the 'search' button, the system shall return search results matching the entered drug name or related keywords.
- FR-6: The patient shall be able to search by a drug's name or drug's category.
- FR-7: The patient shall be able to filter available drugs (with no search) by nearest pharmacies and price range.
- FR-8: After hitting the search button, the patient shall see analytics of the drug's average price, its highest and lowest price, and its availability percentage in pharmacies.
- FR-9: The system shall suggest alternative drugs if the searched drug is available in fewer than 3 pharmacies.
- FR-10: The patient shall be able to:
 - rate pharmacies.
- FR-11: The patient shall be able to find all available pharmacies which have a specific drug. And see a pharmacy's profile which contains:
 - Pharmacy logo.
 - Location.
 - Rate.
 - Available drugs in it.
 - Administrator Doctor.
 - Contact number.

- FR-12: The patient shall be able to mark a drug as a favorite to get any updates related to the drug, like:
 - Change in price.
 - Availability in new pharmacies.
 - Available alternatives.
- FR-13: The system shall notify a pharmacy if a drug frequently searched by patients in its area is currently unavailable in that pharmacy.
- FR-14: For the pharmacy's inventory in the system, the pharmacy should be able to update (insert, modify, delete) the inventory manually (using the system's UI itself), or by linking with its database itself using API (refreshed daily or manually – the pharmacy can freely choose between these two methods).
- FR-15: A pharmacy shall be able to see analytics about:
 - traffic on their page.
 - Traffic on the different drugs to find trends to optimize their stack of drugs.
- FR-16: Admins shall be able to:
 - track traffic on drugs and pharmacies.
 - Monitor database activity.
 - Generate reports on different activities on the system.
- FR-17: Data must be refreshed daily.
- FR-18: If the user did not find any results, his search must be preserved until the required drug become available again or its alternatives, then the patient should be notified that what he needs become already available now.

Non-functional requirements

- **Performance:** The system shall respond to search queries within 5 seconds under normal load conditions.
- **Security:**
 - passwords must be encrypted.
 - Role-based access control must be implemented.
- **Usability:** the system must support English and Arabic languages.
- **Reliability:** System uptime shall be at least 95% monthly.

Assumptions

- Pharmacies will regularly update drug availability.
- Patients have internet access.
- Drug data is provided and maintained by pharmacies.